

Auteur: Rob Willemsen
 Studierichting: Informatica
 Oefeningenreeks 2D nr. 10.14

$$\begin{aligned} & \lim_{x \rightarrow 2} \left(\frac{x^3 - x^2 - 8x + 12}{3x^3 - 10x^2 + 4x + 8} \right) \\ &= \frac{2^3 - 2^2 - 8(2) + 12}{3(2)^3 - 10(2)^2 + 4(2) + 8} \\ &= \frac{8 - 4 - 16 + 12}{24 - 40 + 8 + 8} = \frac{0}{0} \end{aligned}$$

Teller:

$$\begin{array}{c|ccc} & 1 & -1 & -8 & 12 \\ 2 & & 2 & 2 & -12 \\ \hline & 1 & 1 & -6 & 0 \\ 2 & & 2 & 6 & \\ \hline & 1 & 3 & & 0 \end{array}$$

Noemer:

$$\begin{array}{c|ccc} & 1 & -1 & -8 & 12 \\ 2 & & 2 & 2 & -12 \\ \hline & 1 & 1 & -6 & 0 \\ 2 & & 2 & 6 & \\ \hline & 1 & 3 & & 0 \end{array}$$

$$\lim_{x \rightarrow 2} \left(\frac{(x-2) \cdot (1x+3)}{(x-2) \cdot (3x+2)} \right)$$

$$\lim_{x \rightarrow 2} \left(\frac{1x+3}{3x+2} \right)$$

$$= \frac{1(2)+3}{3(2)+2}$$

$$= \frac{5}{8}$$

References